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Assessment of five global gridded precipitation estimates over a southern Mediterranean basin (Tensift, Morocco)

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ABSTRACT

Gridded Precipitation Products (GPPs) could exhibit discrepancies related to detecting precipitation amounts and patterns. This paper aims to evaluate the accuracy of five GPPs currently in operational production over the Tensift basin in the southern Mediterranean. The GPPs are reanalysis-based (ERA5, ERA5-Land, MERRA-2) and multi-source data fusion (TerraClimate, MSWEPv2.8). Their annual and monthly precipitation are compared to observations from fourteen ground gauges over the entire period from 1980 to 2021 and each decade of this period. A set of statistical metrics, such as the Kling Gupta Efficiency (KGE) and the Root Mean Square Error (RMSE), as well as the Bias, served to carry out the evaluation. Four main findings could be highlighted: (i) GPPs have, in general, a good correlation with gauge data; hence they could be used to study the temporal variability of the observed annual and monthly precipitation. (ii) ERA5-Land does not bring significant improvements in precipitation estimates apart from its finer spatial resolution. (iii) GPPs perform better in the plain than in the mountains. (iv) TerraClimate and MSWEPv2.8 present better consistency across the decades. ERA5 and TerraClimate, the longest data series, were used to visualize the precipitation trends in the basin.

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1. Introduction

Accurate precipitation data is vital for climate monitoring and hydrological studies and for assessing water availability for agriculture and drinking purposes. Reliable ground-based precipitation is difficult to collect, especially over a large basin. The first reason is linked to precipitation systems that could be erratic and heterogeneous in time and space (Los et al. 2001; Sun et al. 2014). The second reason is that the

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ground precipitation measuring instruments are not always sufficient, scarcer in high topography and remote areas, and not always well and continuously measured (Liu et al. 2013; Sun et al. 2018). To bridge the gap in the availability of precipitation data over large areas, the global gridded precipitation products (GPPs) have made great strides in recent decades, providing a significant amount of continuous short- and long-term data, at different temporal and spatial scales, for regional and global precipitation monitoring (Amjad et al. 2020; Valencia et al. 2023).

The reanalysis products are currently among the widely used GPPs. They combine a fixed data assimilation system with global observations and satellite data, enabling the quantification of multiple fluxes and processes in the coupled earth-atmosphere system (Zhu et al. 2021; Jiang et al. 2024). Examples of state-of-the-art reanalysis products are JRA-55 (the Japanese 55-year Reanalysis) conducted by the Japan Meteorological Agency (JMA) (Kobayashi et al. 2015), ERA5, and ERA5-Land provided by the European Center for Medium-Range Weather Forecasts (ECMWF) (Hersbach et al. 2020), the Modern-Era Retrospective Analysis for Research and Applications (MERRA) developed by The Global Modeling and Assimilation Office (GMAO) (Rienecker et al. 2011). MERRA-2 was released to enhance the original MERRA dataset. It has recently improved atmospheric assimilation, allowing it to employ more current satellite measurements that the original MERRA system was unable to include (Rienecker et al. 2011; Gelaro et al. 2017).

Other emerging GPPs are based on multi-source precipitation data fusion, that could combine data from gauges, satellites, and reanalysis. MSWEPv2.8 (the Multi-Source Weighted-Ensemble Precipitation) is based on the combination of daily gauge data, satellite, and reanalysis precipitation estimates to improve performance in densely gauged, convection-dominated, and frontal-dominated regions (Beck et al. 2017, 2019). TerraClimate is another product that employs a multisource interpolation technique that combines the high spatial resolution of WorldClim with the fine-grained temporal resolution of Climatic Research Unit Time series (CRU Ts4.0) and the comprehensive 55-year Japanese reanalysis (JRA55) (Abatzoglou et al. 2018).

GPPs may exhibit discrepancies in the magnitude and variability of precipitation patterns, which vary across the regions and the basins of the globe (Sun et al. 2018; Hu et al. 2024). This explains the continuous efforts made to release new precipitation estimates from advanced processing and consistent data to gain better accuracy, finer resolutions (in time and space), and longer temporal coverage. Evaluation studies of those products are necessary both at global, regional, and basin scales, in search of the best product for a given area or a specific application. Evaluation studies would also help identify product failures and weaknesses, necessary to orient future improvements.

In the present study, five GPPs are selected from the above-mentioned products to assess their performance over a large southern Mediterranean basin with a complex topography and an arid to semiarid climate. We analyzed datasets relying on reanalysis (ERA5, ERA5-Land, and MERRA-2), and multisource fusion products (MSWEPv2.8, TerraClimate). To date, the reanalysis (ERA5, ERA5Land) and the (MERRA-2) are known for being state-of-the-art precipitation datasets. The accuracy of these products has been evaluated at global (Lavers et al. 2022) and regional scales

(Liu et al. 2018; An et al. 2020; Xu et al. 2022a). Chen et al. (2021), Gomis-Cebolla et al. (2023), and Jiao et al. (2021) evaluated the data performance of ECMWF-derived reanalyzes in Nepal, China, and Spain respectively. These studies concluded that reanalyzes capture spatial patterns in observed precipitation. MERRA-2 was widely used for multiple applications and revealed that it recorded precipitation amounts, but has not identified extreme precipitation events in some regions (Reichle et al. 2017; Hamal et al. 2020; Arshad et al. 2021; Song and Wei 2021; Xu et al. 2022b).

The evaluation of the GPPs over the basin is undertaken for annual and monthly precipitation, using a set of precipitation gauge data from different altitudes. The monthly resolution of this study corresponds to agroclimatic concerns such as drought. The performance is analyzed for the entire study period, from 1980 to 2021, as well as for each of the four decades of this period.

The main objectives of the study are (i) The assessment of the performance of the GPPs over 41 years, compatible with the needs of climatic and hydrological studies. (ii) The analysis of the variation of their performance between plains and mountains. (iii) The analysis of the consistency of the products over decades. (iv) The use of the longest data sets to visualize the long-term trends in the basin.

2. Data and methods

2.1. Study area

Morocco, situated on the southern edges of the path of mid-latitude frontal storm systems that cross the North Atlantic, experiences relatively low levels of precipitation, which gradually decrease from north to south (Driouech et al. 2010). The precipitations vary by season, and a Mediterranean climate, with moderate, wet winters and hot, dry summers characterizes the regions along the coast. During the rainy season, which generally extends from October to April, ridges of high pressure sometimes form offshore, diverting storms northwards. The prolonged presence of these ridges leads to drought conditions. In addition, the cold Canary Current along the West Coast reinforces atmospheric stability, reducing the likelihood of precipitation (Zamrane et al. 2016). May to September is dry, and the limited precipitations are mostly convective.

The Tensift basin in central Morocco (7.17°–9.38° W and 30.85°–32.15° N; 24000 km^2) shows a heterogeneous orography (Figure 1), comprising the High-Atlas mountains (altitude from 1000 m to 4167 m), the piedmont (altitude from 600 m and 1000 m), and the Haouz plain (altitude from 150 m to 600 m). Due to the influence of the topography, the climate shows spatial variability according to the Köppen-Geiger climate classification map. The climatic context is characterized by a dominant semi-arid climate (BSh, BSk). An intersection between Mediterranean (Csa) and cold (Dsa) climates with dry summers governed by the high altitudes of the basin. The elevation has a strong influence on the precipitation pattern, with precipitation amounts significantly higher in the mountains (up to 500 mm/year) and low in the plain (less than 200 mm/year). The High-Atlas mountains create significant orographic precipitation, giving rise to a network of wadis that cross the region, joining to form wadi Tensift,

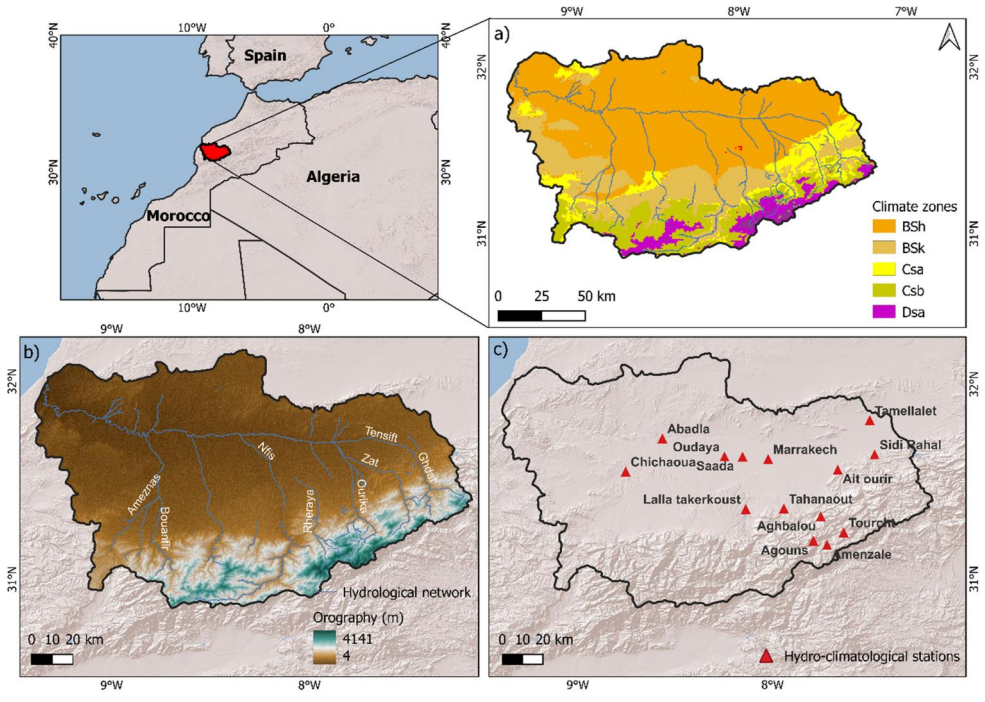


Figure 1. a) The Tensift basin with the Köppen-Geiger climate classification zones (Beck et al. 2018), b) Digital elevation map (DEM) of the study area, c) spatial distribution of rain-gauges.

which discharges into the Atlantic Ocean. The streamflow is derived or stored in dams to supply irrigated agriculture in the plain. The basin faces serious problems of drought (Oukaddour et al. 2024), water scarcity, and water resource mismanagement (Le Page et al. 2012).

The Tensift basin is also a well-known environmental observatory (https://www.lmi-trema.ma/) that has progressively evolved since 2002 (Chehbouni et al. 2008) toward a comprehensive hydro-meteorological observatory (Jarlan et al. 2015) benefiting from a relatively large network of precipitation gauges.

2.2. Dataset

2.2.1. Reference precipitation observations

In situ precipitation data for the Tensift basin were provided by ABHT (Agence du Bassin Hydraulique du Tensift), and ORMVAH (Office Régional de Mise en Valeur Agricole du Haouz). The total number of stations used is 14 (4 in the mountains, 4 in the piedmont, and 6 in the plains). In this work, we have been considering three geographical zones: the plain (altitude below 600 m), the Atlas foothills (altitude between 600 and 1000 m), and the mountains (altitude above 1000 m). The records from the mountain stations cover a period of 23 years (1998–2021), while the rest of the stations were used for a historical period of 41 years (1980–2021). We have chosen the common period for most valid stations in the basin (Table 1). Data are collected in daily and monthly time steps. Daily data were aggregated to obtain the

Table 1. Weather stations characteristics.

Location	Gauge	Latitude	Longitude	Altitude (m)	Period
Mountain	Agouns	31.19	−7.79	2381	1998–2021
	Amenzale	31.17	−7.72	2374	1998–2021
	Tourcht	31.24	−7.63	1615	1998–2021
	Aghbalou	31.32	−7.75	1070	1981–2021
Piedmont	Tahanaout	31.36	−7.94	925	1981–2021
	Ait Ourir	31.56	−7.66	760	1981–2021
	Sidi Rahal	31.64	−7.47	695	1981–2021
	Lalla Takerkoust	31.36	−8.13	614	1981–2021
Plain	Tamellalet	31.81	−7.50	588	1981–2021
	Marrakech	31.62	−8.02	457	1981–2021
	Saada	31.63	−8.15	415	1981–2021
	Oudaya	31.63	−8.25	377	1981–2021
	Chichaoua	31.55	−8.75	340	1981–2021
	Abadla	31.72	−8.56	263	1981–2021

sum of precipitation per month. The selected time series were checked and processed to ensure no missing values. The final choice of 14 stations was based on data availability and the objective of obtaining the longest possible continuous records for reliable analysis.

2.2.2. Precipitation products

A summary of the products considered in the present study is listed in Table 2. The selection of precipitation products (ERA5, ERA5-Land, MERRA-2, TerraClimate, and MSWEPv2.8) was guided by several key criteria. They are currently in operational production, guaranteeing future data availability, as well as their long-term coverage (at least starting from 1980) which is essential for climate trend analysis. In addition, they have different spatial (40 km to 5 km) and temporal resolutions (hourly to monthly) to capture differences in precipitation patterns and dynamics and widespread use in similar hydro-climatic studies, underlining their reliability and scientific relevance. In addition, ERA5-Land, MERRA-2, TerraClimate, and the newest MSWEPv2.8 have not been tested over the whole Tensift basin. ERA5 was compared to satellite precipitation data in two previous studies, one for daily precipitation over the period 2007 to 2019 (Salih et al. 2022), and the second for monthly precipitation over the period 1983 to 2017 (Najmi et al. 2023), making their assessment particularly useful for understanding precipitation variability and data performance.

The distribution of stations and the grid cells of the products are shown in Figure 2, along with the annual average precipitation map. Spatial interpolation was carried out using the inverse distance weighting (IDW) method to obtain the map from the observed data. IDW relies on the principle that nearby points influence the predicted values more than distant ones, based on the inverse relationship between distance and similarity. This method calculates a weighted average of known values, with weights inversely proportional to the distance from the unknown location and their sum equal to one (Camera et al. 2014; Workneh et al. 2024).

2.2.2.1. ERA5 and ERA5-Land. ERA5 is the latest climate reanalysis developed by the ECMWF. It provides an accurate view of the evolution of land variables at high temporal resolution (1 h) based on the Integrated Forecasting System (IFS) Cy41r2

Table 2. Summary of the used GPPs products.

Dataset	Resolution		Coverage		Websites
	Spatial	Temporal (Finest)	Spatial	Temporal	
TerraClimate	0.04° × 0.04°	Monthly	Global	1958 to near-present	https://www.climatologylab.org/terraclimate.html
ERA5-Land	0.1° × 0.1°	Hourly	Global	1950-present	https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-land-monthly-means?tab=overview
ERA5	0.25° × 0.25°	Hourly	Global	1940-present	https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels-monthly-means?tab=overview
MERRA-2	0.5° × 0.625°	Daily	Global	1980-present	https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/
MSWEP	0.1°× 0.1°	3-hourly	Global	1979 to near real-time	https://www.gloh2o.org/mswep/

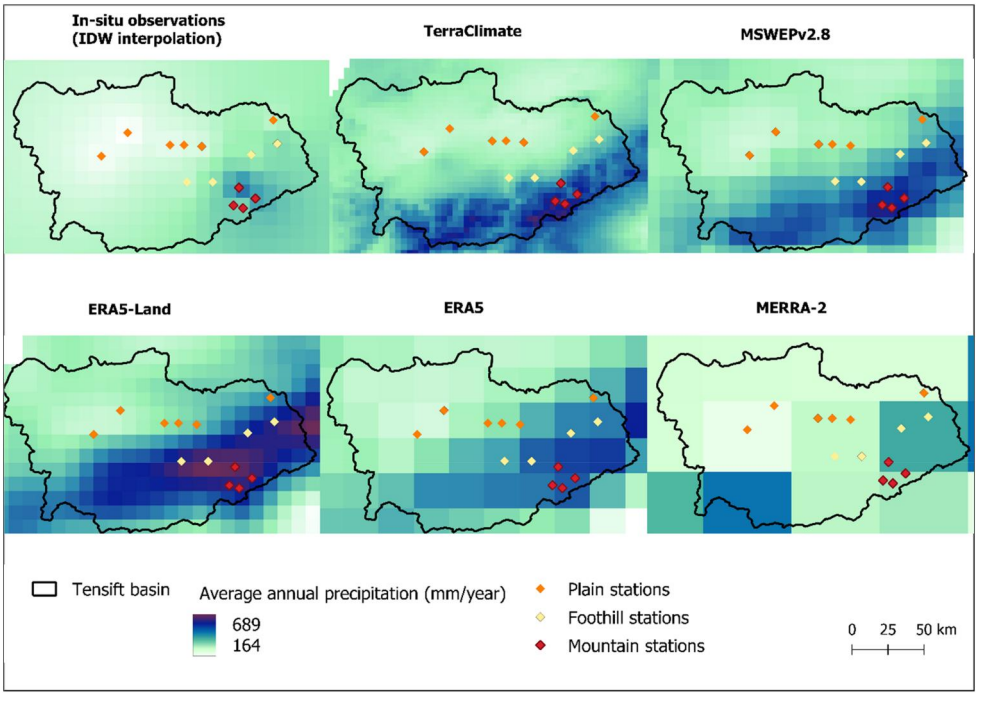


Figure 2. Mean annual precipitation (mm/year) in observations, TerraClimate, MSWEPv2.8, ERA5-Land, ERA5, and MERRA-2 during 1981–2021.

(Hersbach et al. 2020). ERA5 is produced using a 4D-Var scheme, integrating measurements from various observational systems. Combining the atmospheric component of the IFS with the land-surface model HTESSEL (Revised Tiled ECMWF Scheme for Surface Exchanges over Land) and the ocean wave model (WAM) further offers a physically relevant analysis of the land surface and ocean.

The ERA5-Land reanalysis (9 km) (Muñoz-Sabater et al. 2021) is a spatially enhanced version of the ERA5 reanalysis (31 km). To estimate ERA5-Land, air

temperature, air humidity and pressure from ERA5 have been corrected for the altitude difference between the forcing grid and the higher resolution grid of ERA5-Land, using the 'lapse rate correction' (Dutra et al. 2020). The monthly total precipitation (tp) of ERA5 and ERA5-Land were used in this study.

2.2.2.2. MERRA-2. MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2) is created using an updated version of the GEOS (Goddard Earth Observing System Model, version 5.12.4) atmospheric data assimilation system. MERRA-2 precipitation features several major advances including the use of a merged satellite-gauge precipitation product to correct the precipitation falling on the land surface instead of the gauge-only data used for MERRA (Reichle et al. 2017). The MERRA-2 (Total surface precipitation (M2TMNXFLX_5.12.4)) incorporates data from new microwave sounders, hyperspectral infrared radiance instruments, as well as other novel data sources, in addition to in-situ and remotely sensed atmospheric observations (Reichle et al. 2017). The spatial resolution of the MERRA-2 data is 0.5° × 0.625° (about 55.5 km × 69.37 km at the Tensift basin latitude). Its temporal resolution is 1 h (Gelaro et al. 2017).

2.2.2.3. MSWEP. MSWEP (Multi-Source Weighted-Ensemble Precipitation) combines multiple data sources, spanning the period 1979 to the present. The datasets come with high-resolution daily estimates of global precipitation updated with a three-hour latency at 0.1° resolution (about 11.1 km at the Tensift latitude). MSWEPv2.8 (Multi-Source Weighted-Ensemble Precipitation, version 2.8) optimally merges gauge, satellite, and reanalysis data, using a weighted ensemble approach. Three different variants of MSWEPv2.8 are available. The 'Past_nogauge' variant (1979–2020) integrates data from CHELSA, ERA5, IMERG (post-2000), and GridSat (pre-2000). The 'Past' variant (1979–present) incorporates daily gauge corrections from GHCN-D, GSOD, and some national gauge databases, and the 'NRT' variant extends 'Past_nogauge' recording to the present with a latency of around 3 h, and is based on ERA5, GDAS, GSMaP, and IMERG (Beck et al. 2019). The 'Past_nogauge' variant is used and extended in the present study using the 'NRT' data until 2021. This precipitation product is the only one to take into account gauge reporting times, which is essential to minimize temporal discrepancies between satellite/reanalysis data and gauge observations.

2.2.2.4. TerraClimate. TerraClimate uses a set of high-precision monthly climate data to generate global estimates of precipitation across terrestrial surfaces at a high spatial resolution (1/24°, ∼4 km). A machine learning approach is used to produce reliable precipitation estimates by integrating information from a variety of sources. High-spatial-resolution climate normals from WorldClim (Fick and Hijmans 2017), coarse-resolution time-varying data from the Climate Research Unit - CRU (Harris et al. 2014) and the Japanese 55-year reanalysis - JRA-55 (Kobayashi et al. 2015) are combined to produce TerraClimate datasets by climatically aided interpolation.

2.3. Evaluation method and metrics

Statistical measures including correlation coefficient, Root Mean Square Error (RMSE), bias, and Kling–Gupta efficiency (KGE) are typically used to evaluate the accuracy of precipitation products. They have shown their efficiency both in small- (Salih et al. 2022) and large-scales (Saemian et al. 2021; Xu et al. 2022b; Gomis-Cebolla et al. 2023) in low-lands (Polong et al. 2023; Sun et al. 2016) and mountains (Mo et al. 2024; Sun et al. 2023). In this study, the five precipitation products have been evaluated at a point scale against gauge observations. The statistical metrics are briefly described below.

A commonly used measure for evaluating the strength of the linear relationship between simulated and observed data is the Pearson correlation coefficient (r). There is no correlation between the precipitation estimates and the gauges data if $r = 0$, and the optimum value is 1. The closer r is to 1, the better the products reproduce the observed temporal precipitation variability.

$$r = \frac{\sum_{i=1}^{n}(x_i - \overline{x_i})(y_i - \overline{y_i})}{\sqrt{\sum_{i=1}^{n}(x_i - \overline{x_i})^2 \sum_{i=1}^{n}(y_i - \overline{y_i})^2}} \tag{1}$$

Where y_i and x_i denote precipitation amounts from products and rain gauges, respectively, $\overline{x_i}$ and $\overline{y_i}$ are the averages of observed and product-derived data.

The root mean square error (RMSE) is one of the commonly used error indexes in statistics. RMSE measures the average difference between gauge data and precipitation estimates. The lower the RMSE value, the more accurate the precipitation products.

$$RMSE = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \tag{2}$$

The Bias measures the systematic difference between the observed and simulated precipitation datasets. It is a statistical measure of the error-index. Negative bias indicates that a product underestimates the precipitation, and inversely.

$$BIAS = \frac{\sum_{i=1}^{n}(y_i - x_i)}{\sum_{i=1}^{n}(x_i)} \tag{3}$$

The Kling–Gupta efficiency (KGE) (Gupta et al. 2009) is obtained by the decomposition of Nash–Sutcliffe efficiency (NSE) into three equally weighted constituent components (correlation, variability ratio, and mean bias ratio). Thus, KGE assesses the ability of the precipitation products to reproduce temporal variability and to estimate precipitation means. The best values of KGE go toward 1.

$$KGE = 1 - \sqrt{(r-1)^2 + \left(\frac{\sigma_S}{\sigma_{obs}} - 1\right)^2 + \left(\frac{\mu_S}{\mu_{obs}} - 1\right)^2} \tag{4}$$

Where r is the linear correlation between observations and simulations, σ_{obs} is the standard deviation in observations, σ_S the standard deviation in simulations, μ_S the simulations data mean, and μ_{obs} the observations mean.

When evaluating satellite precipitation products or gridded precipitation datasets against rain gauge observations, KGE helps identify whether

Captures precipitation timing (correlation component) - Does the model predict rain events when they actually occur?

Reproduces precipitation variability (variability component) - Does the model show similar ranges from light to heavy rainfall as observed?

Estimates precipitation amounts correctly (bias component) - Does the model systematically over- or under-estimate rainfall amounts?

2.4. Long-term trend analysis

We used the non-parametric statistical tests of Mann-Kendall (Kendall 1938; Mann 1945) and the Theil-Sen slope estimator (Theil 1950; Sen 1968) to detect significant monotonic trends in precipitation time series. The time series considered here are the longest time series available, including ERA5 reanalysis and TerraClimate. The analysis was carried out over different periods in order to study the variability of precipitation patterns and their influence on trends.

Change point detection was performed using Kernel density estimation (Harchaoui and Cappe 2007), a robust non-parametric model that detects multiple change points in time series based on mean, and variance. It can go beyond these two parameters to find more complex change points.

3. Results

3.1. Performance of the annual precipitation estimates

The results of the evaluation of the annual precipitation using statistical metrics are displayed in Figure 3. The reference stations were positioned in order of altitude, starting with the Agouns station at 2381 m altitude, down to the Abadla station at 250 m altitude.

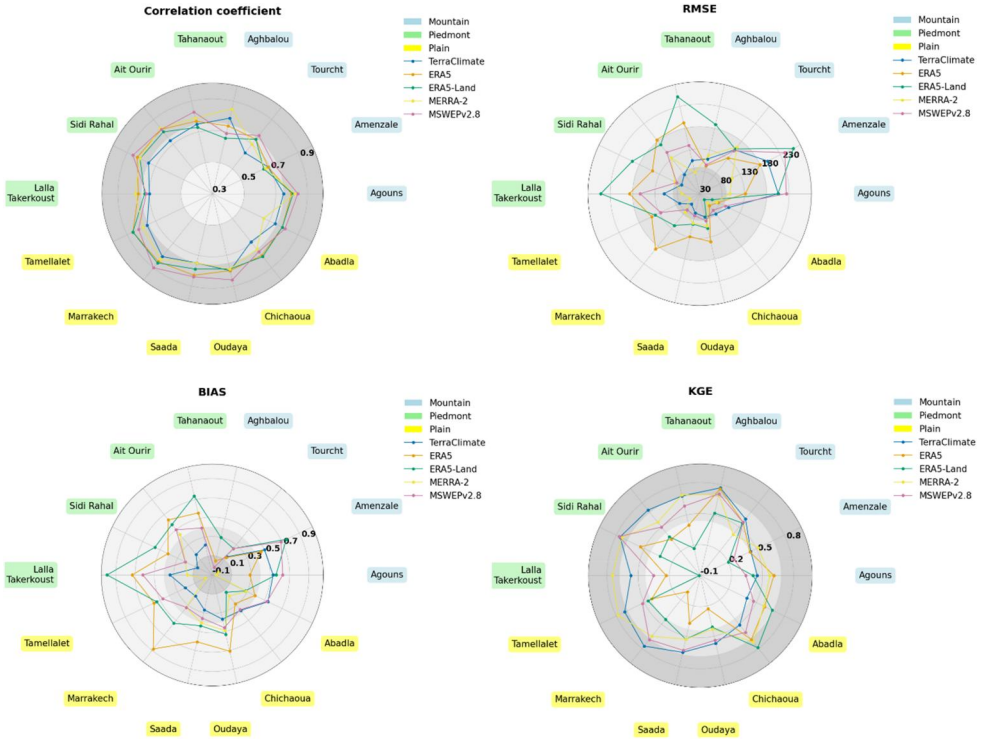


Figure 3. Plots for performance metrics of TerraClimate, MSWEPv2.8, MERRA-2, ERA5, and ERA5-Land with observed datasets for annual timescale, the darker zones indicate areas with the best GPP performance.

Over the studied periods - (1998–2021) for the Agouns, Amenzale, and Tourcht stations in the mountains and (1980–2021) for the rest of the stations - the precipitation estimates generally showed good agreement with gauge observations, as indicated by the correlation coefficient ranging from 0.63 to 0.9. Thus, the products could reproduce, at different degrees the interannual variability of the precipitation. MSWEPv2.8, ERA5, and ERA5-Land show the highest correlation values. For MSWEPv2.8, r varies from 0.7 to 0.9 and is above 0.8 for 9 of 14 total gauges.

In terms of accuracy, MERRA-2 exhibits the lowest RMSE and bias with high-altitude gauges. TerraClimate and MSWEPv2.8 consistently exhibit low RMSE and bias values across most stations, except the most elevated ones (Amenzale and Agouns). The ERA5 products have the lowest accuracy, the KGE values corroborate their lower performance. Among the precipitation products analyzed, only TerraClimate and MERRA-2 estimates have a good performance in terms of KGE at the stations located at the highest altitudes. MSWEPv2.8 then outperforms MERRA-2 at low-altitude stations and shows higher KGE values.

Those features are observed in the interannual precipitation variability over the Tensift basin (Figure 4). The observed mean annual precipitation is around 200 mm in the Abadla station and 500 mm in the high-altitude stations. The products generally fit the variability of the observed annual precipitation. The ERA5, ERA5-Land, and MSWEPv2.8 show an overestimation of annual precipitations, which increases in high altitudes. TerraClimate loses its precision in high mountains. MERRA-2 presents relatively consistent annual precipitation variability.

3.2. Performance of the monthly precipitation estimates

The evaluation of monthly precipitation estimates against ground observations reveals that generally, the correlations remain good, although deviations are observed at the high mountain stations (Tourcht, Agouns, and Amenzale) where r varies between 0.51 and 0.77 at the monthly scale (Figure 5). In the plain, the correlation with monthly precipitation even has improved compared to annual precipitation, especially for ERA5, ERA5-Land, and TerraClimate.

The RMSE shows that the products have similar accuracy, except for ERA5-Land, which has the largest discrepancies. The bias separates MERRA-2 and TerraClimate from MSWEPv2.8, ERA5, and ERA5-Land, which tend to overestimate data. The KGE indicates good performance by MERRA-2 and TerraClimate in the plain, with MERRA-2 outperforming in the mountains. MSWEPv2.8 performs notably well in the Piedmont and plain stations compared to the other datasets.

Monthly precipitation averages were calculated over 41 years (1980–2021) for piedmont and lowland stations, and over 23 years (1998–2021) for mountain stations (Figure 6). Precipitation estimates for the four seasons may be considered consistent for the plain stations. However, the discrepancies increase as the altitudes become higher. Indeed, for the mountains ERA5 and ERA5-Land overestimate the low precipitation of summer (June, July, August) and autumn (September, October, November).

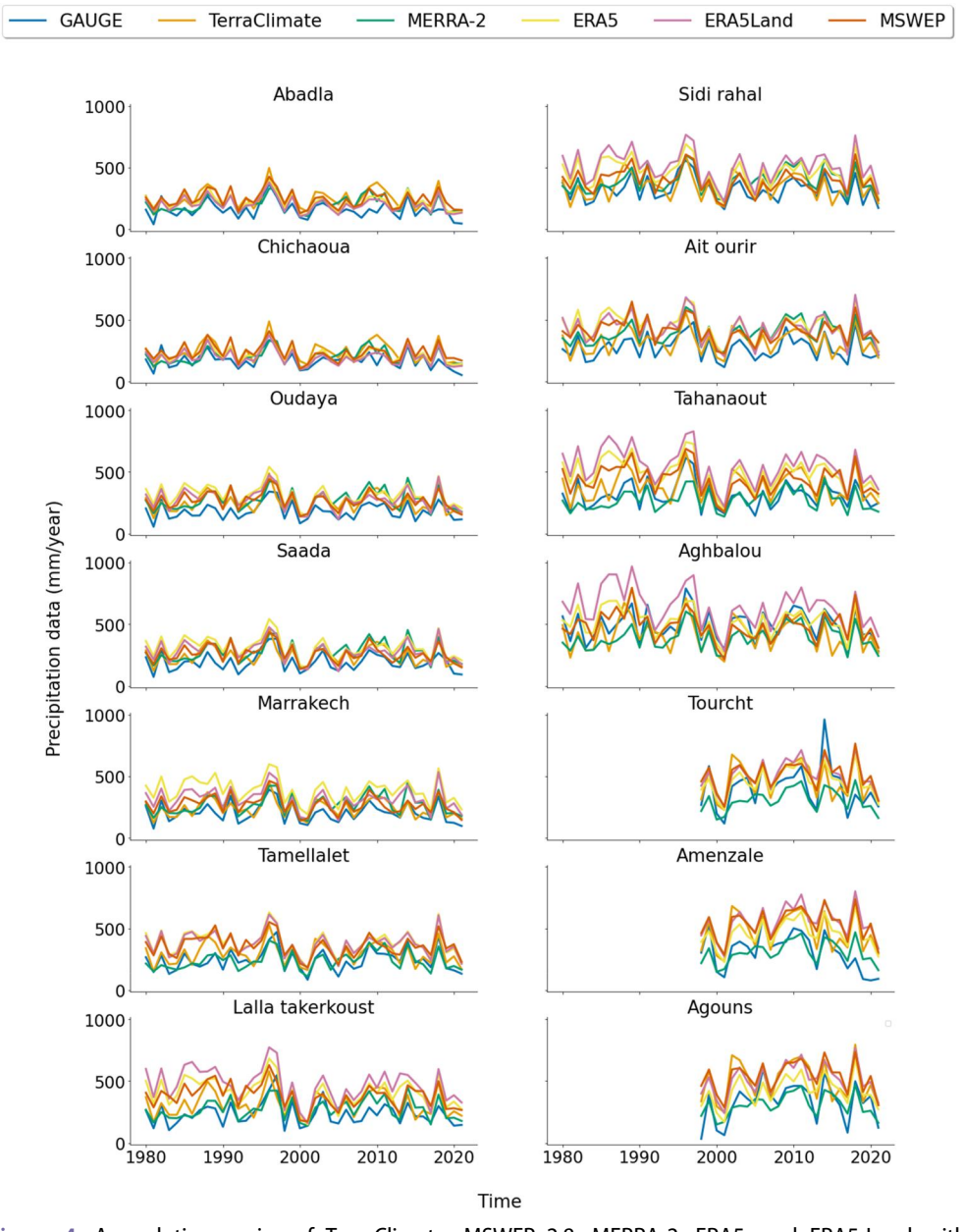


Figure 4. Annual time series of TerraClimate, MSWEPv2.8, MERRA-2, ERA5, and ERA5-Land with in situ precipitation data over the Tensift basin.

MERRA-2 is closer to the observed data, but it poorly represents their seasonality as it underestimates the precipitation of winter (December, January, February) and spring (March, April, Mai). TerraClimate and MSWEPv2.8 display a clear seasonality, but overestimate the precipitation peaks in autumn and winter.

The same analysis was conducted over the common period from 1998 to 2021 (Figures A1–A4, Supplementary Material). The results indicate minor differences in

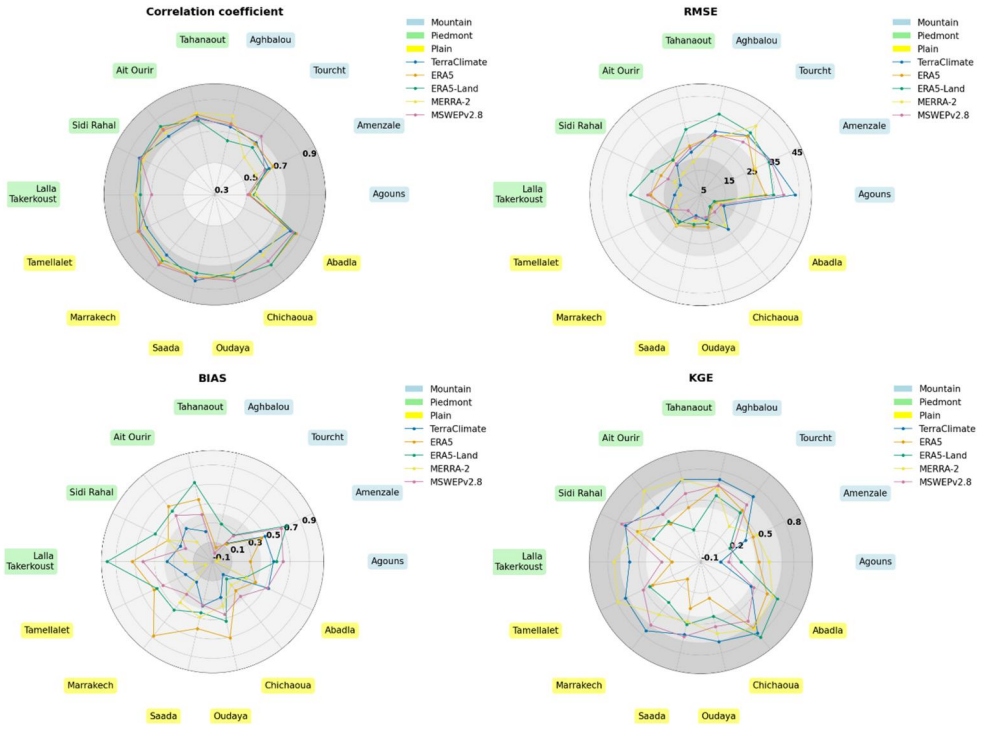


Figure 5. Plots for performance metrics of TerraClimate, MSWEPv2.8, MERRA-2, ERA5, and ERA5-Land with observed datasets for monthly timescale, the darker zones indicate areas with the best GPP performance.

performance at some stations, while the majority of metrics remained within the same range of variation. The TerraClimate dataset demonstrated improved performance at the annual and monthly scale across some stations, as demonstrated by higher correlation coefficients at Chichaoua, Oudaya, Saada, Marrakech, and Ait Ourir. There are also lower RMSE values at Aghbalou, Ait Ourir, Marrakech, Oudaya, and Sidi Rahal. These results confirm that the overall conclusions remain consistent even if there is a limitation of a long time series of mountain observations.

3.3. Decadal analysis

To analyze the consistency of the five GPPs over time, the statistical metrics were calculated for each of the four decades (1980–1989, 1990–1999, 2000–2009, 2010–2021) of the study period. The metrics are applied to the monthly precipitations.

The correlation coefficient presents a slight variation between decades compared to the other metrics (Figure 7a). RMSE increases over the decades, except for MSWEPv2.8 (Figure 7b). The highest increase is seen for ERA5 and ERA5-Land for the 2010–2021 decade. The largest dispersions in the bias are recorded for the reanalysis products ERA5, MERRA-2, and ERA5-Land (Figure 7c). During the first decade, 1980–1989, ERA5 and ERA5-Land showed the highest bias, with a median of 0.89 and 0.75 respectively. It decreased during the following decades but remained higher than the other products. Regarding the KGE, the lowest performance, linked

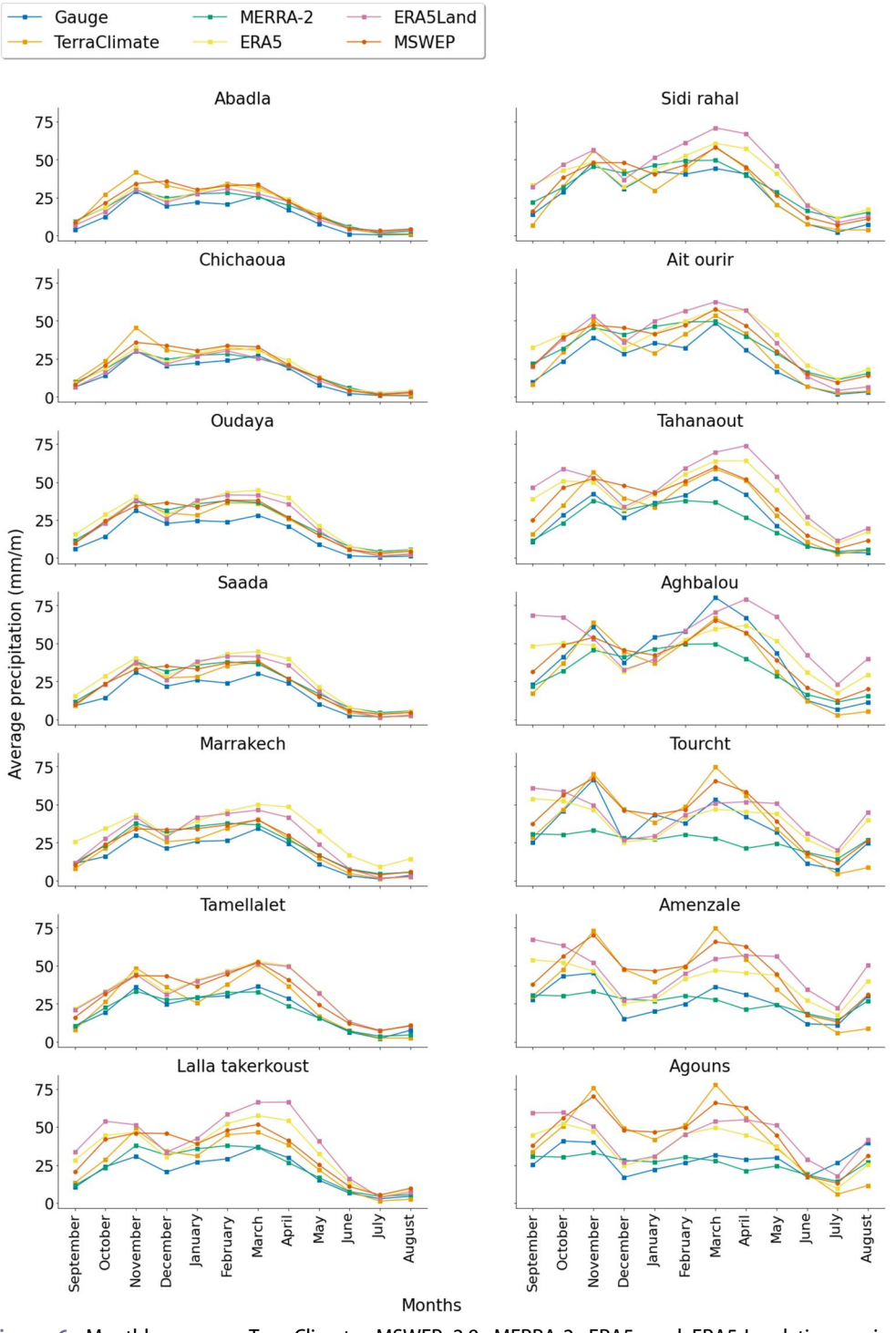


Figure 6. Monthly average TerraClimate, MSWEPv2.8, MERRA-2, ERA5, and ERA5-Land time series with in situ precipitation data over the Tensift basin.

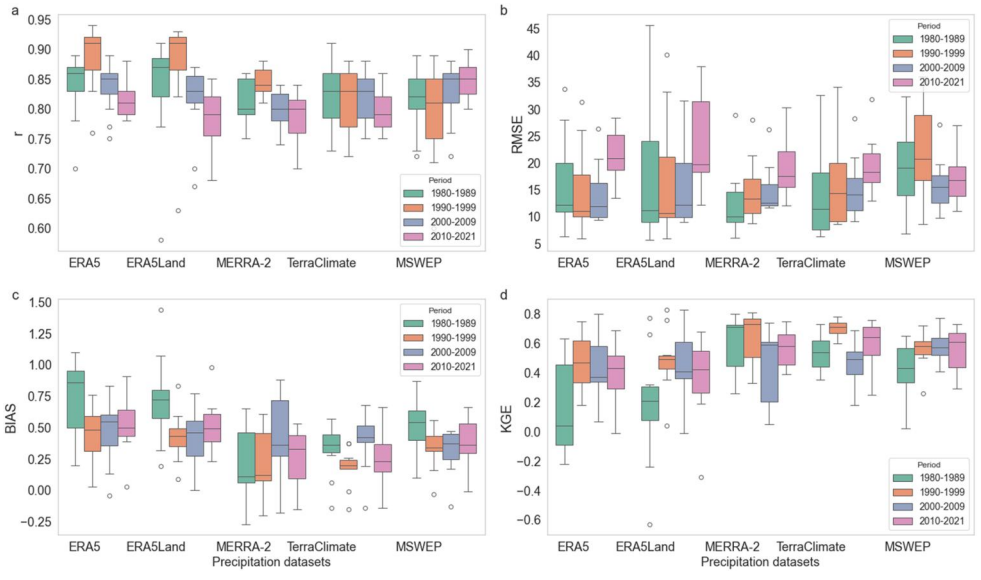


Figure 7. Boxplots for performance metrics of TerraClimate, MSWEPv2.8, MERRA-2, ERA5, and ERA5-Land with observed datasets for monthly timescale.

to the ability to reproduce temporal variability and to estimate precipitation means, is presented by ERA5 during 1980–1989 (Figure 7d) because of their high bias during this decade. TerraClimate and MSWEPv2.8 present the best consistency over time.

3.4. Long-term annual precipitation variability

We analyzed long-term precipitation patterns at three representative stations, Agouns in the mountain, Aghbalou in the Piedmont, and Oudaya in the plain, using ERA5 reanalysis data from 1940 to 2021. The analysis revealed distinct temporal evolution patterns and identified change points in the time series. Overall, the results show no significant trend in precipitation for the whole time series, but segmenting the series at the change points revealed significant variations (Figure 8). From 1940 to 1965, there was a statistically significant increasing trend in precipitation (p-value = 0.04). From 1966 to 1998, the second segment recorded the highest annual precipitations followed by a noticeable decline. The third segment post-1998 shows an increasing slope, with no significant trend, with lower precipitation compared to the previous period.

The change point between 1965 and 1967 was consistently observed across all stations in the ERA5 time series. The trend analysis, combined with change point detection, highlights the influence of the 1966–1998 period on the overall trend, masking the decline in precipitation in the later years. Three precipitation phases can be distinguished. The wet-to-normal phase from the 1940s to the 1970s, was characterized by a rising precipitation trend. This phase is followed by a long period of declining precipitation, reaching a minimum in the 2000s. A slight increase in precipitations is detected after the long dry period.

The analysis of TerraClimate data from 1959 to 2021 shows an overall decline in annual precipitation. Change points were detected in different years for the stations: the

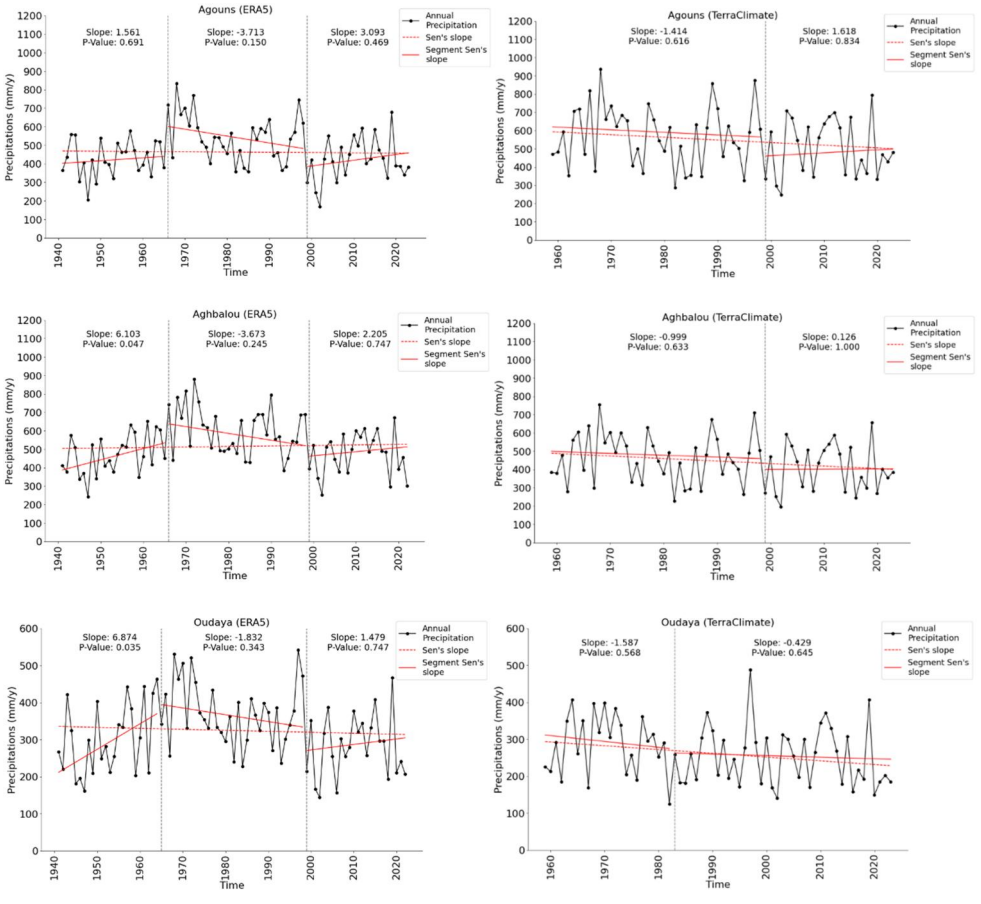


Figure 8. Trends of the annual precipitation estimates derived from ERA5, and TerraClimate over the Tensift basin.

1999 change point is specific to stations of high altitude (Agouns, and Aghbalou), and the 1982 change point is evident at all stations (not shown here). Segments following these change points display a decreasing trend, with a slight reduction in precipitation.

4. Discussion

4.1. Precipitation products performance

In the present study, we analyzed the performance of three reanalysis-based precipitation products: ERA5, ERA5-Land, and MERRA-2, and two multisource fusion products: TerraClimate and MSWEPv2.8. The products are tested over an arid to semiarid climate basin from the Southern Mediterranean. The results showed that the GPPs generally correlate with gauge data, thus exhibiting abilities to reproduce the temporal variability of the observed annual and monthly precipitation in the basin.

In terms of accuracy, the lowest average difference with gauge data (RSME) and the lowest bias are shown by MERRA-2 and TerraClimate. Among the most sought-after products in climatology and hydrology are accurate GPPs with high temporal

and spatial resolutions, and long coverage periods. With those criteria, TerraClimate (4 km, 1958 to present) could be ranked as the best precipitation product for the basin to monitor monthly precipitation. The good performance of TerraClimate was highlighted by various studies that evaluated the product at various spatial scales and under arid and semi-arid climates (de Andrade et al. 2022; Hanchane et al. 2023; Araghi and Adamowski 2024).

The low spatial resolution of MERRA-2 (50 km) restricts its use for small spatial scales. Its low resolution may explain that the product poorly represents precipitation seasonality, underestimating the important precipitations of winter and spring.

Previous studies have shown that while MERRA-2 can reproduce total precipitation amounts on a monthly and annual scale, it often fails to capture intense precipitation events due to its coarse spatial resolution (Reichle et al. 2017; Hamal et al. 2020; Arshad et al. 2021; Song and Wei 2021; Xu et al. 2022b).

The accuracy of ERA5 and ERA5-Land is the lowest of the assessed products. Both products presented the highest bias, thus overestimating precipitation amounts, particularly at high altitudes. This is consistent with previous studies that have evaluated several precipitation datasets in the same arid and semi-arid climatic context (Gomis-Cebolla et al. 2023; Yang et al. 2024). The weaker performance in the mountains compared with the plains of the datasets was reported in Zhang et al. (2024). The issue of overestimating monthly precipitation has been reported in several studies (Liu et al. 2018; Chen et al. 2021; Jiao et al. 2021; Valencia et al. 2023).

ERA5-Land (9 km) as a downscaled version of ERA5 (31 km resolution), was provided by atmospheric forcing of ERA5 downscaled variables. It was corrected using lapse rate correction to account for the elevation effect, due particularly to the overestimation of light and medium precipitations (Pfeifroth et al. 2013). However, as observed in the Tensift basin, ERA5-Land is still less accurate than ERA5. This highlights the importance of considering factors other than spatial resolution alone when selecting precipitation estimates.

MSWEPv2.8 could be ranked in the middle between TerraClimate and MERRA-2 on one side and ERA5 and ERA5-Land products on the other side. MSWEPv2.8 could be used as an alternative to ERA5 precipitation for study periods spanning since 1980, as it intrinsically incorporates ERA5 reanalysis and presents better performance. Numerous studies have reported the high performance of MSWEPv2.8 (Prakash 2019; Valencia et al. 2023; Gupta et al. 2024). This underlines the significance of integrating multiple data sources into a single product. This was demonstrated in the decadal analysis herein, where the MSWEPv2.8 displayed reduced variability, indicating increased long-term consistency for this product.

The problem of the altitude and orography effects is challenging for all the examined products. Our analysis shows clearly that the precipitation estimates deviate from gauge data as moving from the plain to the piedmont and more importantly to the mountains. This problem was previously reported by other studies (Chen et al. 2021; Dubey et al. 2021; Araghi and Adamowski 2024). These discrepancies may be associated with different factors. They could be due to the interpolation methods used for constructing the gridded data, which might not adequately account for the effect of altitude. Furthermore, the resolution of the datasets themselves plays an

important role, particularly in complex terrain where orographic effects lead to more heterogeneous precipitation patterns. It is possible that the ERA5-Land product is being penalized for spatial displacement errors, which are averaged out in lower-resolution products.

The scarcity of observation networks and insufficient monitoring over mountains is also a main reason. For instance, in our basin, a big difference exists between the plain which has many measurement stations with a long temporal coverage, and the mountains which are still under-equipped with measurements that only started late. Finally the ground observations in remote and complex elevated terrains, in addition to their scarcity, could themselves include uncertainties in the magnitude and the variability (Sun et al. 2018).

Another issue demonstrated in the present study is that the performance of the GPPs across the decades may vary. The multisource fusion products, TerraClimate and MSWEPv2.8, present better consistency. The MSWEPv2.8 displayed less variability. For ERA5 reanalysis, the bias was highest between 1980 and 1989. This impairs the overall performance of the products and questions their adequacy for this decade. Thus, precipitation estimates that incorporate the combined advantages of gauge observations with reanalysis and satellite datasets (MSWEPv2.8 and TerraClimate) showed a clear improvement over those that do not.

4.2. Long-term analysis

The analysis of precipitation time series revealed significant variations in trends and breakpoints. These variations are mainly attributed to the coverage period of the used products. The main change points were identified in 1966, 1982, and 1999. Despite differences in temporal coverage of ground gauge data with the ERA5 and TerraClimate datasets, the results agreed on the 1999 change point identified (Figure A5). These results are in line with several climate studies carried out on the basin scale (Ouassanouan et al. 2022; Boughdadi et al. 2023) as well as on larger scales (Born et al. 2010; Driouech et al. 2021). They have shown that the precipitation time series generally shifted in the 1980s. Driouech et al. (2021) highlighted the fact that during this period, precipitation was well below normal, with a deficit of around 40% nationwide. The lowest rainfall was also recorded in 1999–2002, which is a well-known period of severe and prolonged drought in Morocco.

Analysis carried out over different coverage periods makes comparison difficult. However, in general, precipitation patterns show a decrease that was triggered in the late 1960s. This trend has been detected by several studies in Morocco (Khomsi et al. 2016) and other Mediterranean countries (Hoerling et al. 2012; Seager et al. 2019). This period was marked by a drop in precipitation. Another change point in the 1980s was detected in the TerraClimate corresponding to a transitional phase after which precipitation declined in the plain.

Going back a little further in the 1950s, ERA5 data showed that the basin experienced a period with increasing precipitation reaching a peak in 1970. This was found by Sebbar et al. (2011), who analyzed historical rainfall series over the period 1935–2003. They identified two rainfall phases, a normal wet phase beginning in the 1950s and

ending in 1975, and a dry phase following the wet period and lasting until 2003. These variations in climatic trends indicate regional variability influenced by global change (Benassi 2008), highlighting the impact of atmospheric circulation patterns on precipitation variability (Reddaway and Bigg 1996; Xoplaki et al. 2012; Boughdadi et al. 2023).

5. Conclusions

The GPPs tested in our basin present a good correlation with gauge data, thus showing abilities to reproduce the temporal variability of the observed annual and monthly precipitation. Nevertheless, GPPs that incorporate the combined advantages of gauge corrected observations with reanalysis and satellite datasets, MSWEPv2.8 (satellite, and reanalysis) and TerraClimate (geostationary thermal infrared (TIR), gauge, model), showed a clear improvement in precipitation amounts estimate in our basin over the other products. Among the reanalysis, MERRA-2 is the best product. However, its coarse spatial resolution of 50 km is likely to restrict its use for small scale basins and in regions with complex topography that require fine resolution. The reanalysis ERA5 and ERA5-Land are the products that most overestimate the precipitation amounts in the basin. The bias for the first tested decade (1980–1989) is the highest and needs to be corrected to improve the performance of those products. ERA5-Land, which is an improved version of ERA5, does not resolve the precipitation issues of those reanalyzes. In our basin, MSWEPv2.8 could be an alternative to ERA5 precipitation for study periods spanning since 1980, as it intrinsically incorporates ERA5 reanalysis and presents better performance.

Our analysis also showed that the precipitation estimates for the studied products progressively deviate from observational data as moving from the plain to the piedmont, and the differences are more important in the mountains. As mountain precipitation is crucial in assessing and simulating hydrological processes, huge efforts are still needed to equip mountains and improve the monitoring of atmospheric parameters. So far, the GPPs can hardly bridge the gap in the availability of ground precipitation data, underscoring the need for integrating more ground data into the climate models and for effective bias adjustments. Future research could investigate the underlying causes of identified change points using ERA5 and TerraClimate datasets and assess their performance within periods defined by the breakpoints to understand the influence of precipitation regimes.

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Disclosure statement

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Data availability statement

All data are freely available on the data providers' websites: ERA5 and ERA5-Land (https://cds.climate.copernicus.eu/); MERRA-2 (https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/); TerraClimate (https://www.climatologylab.org/terraclimate.html); MSWEPv2.8 (https://www.gloh2o.org/mswep/).

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